NATIONAL COMPOSITES CENTRE

CATAPULT High Value Manufacturing

Aircraft Track Fairing

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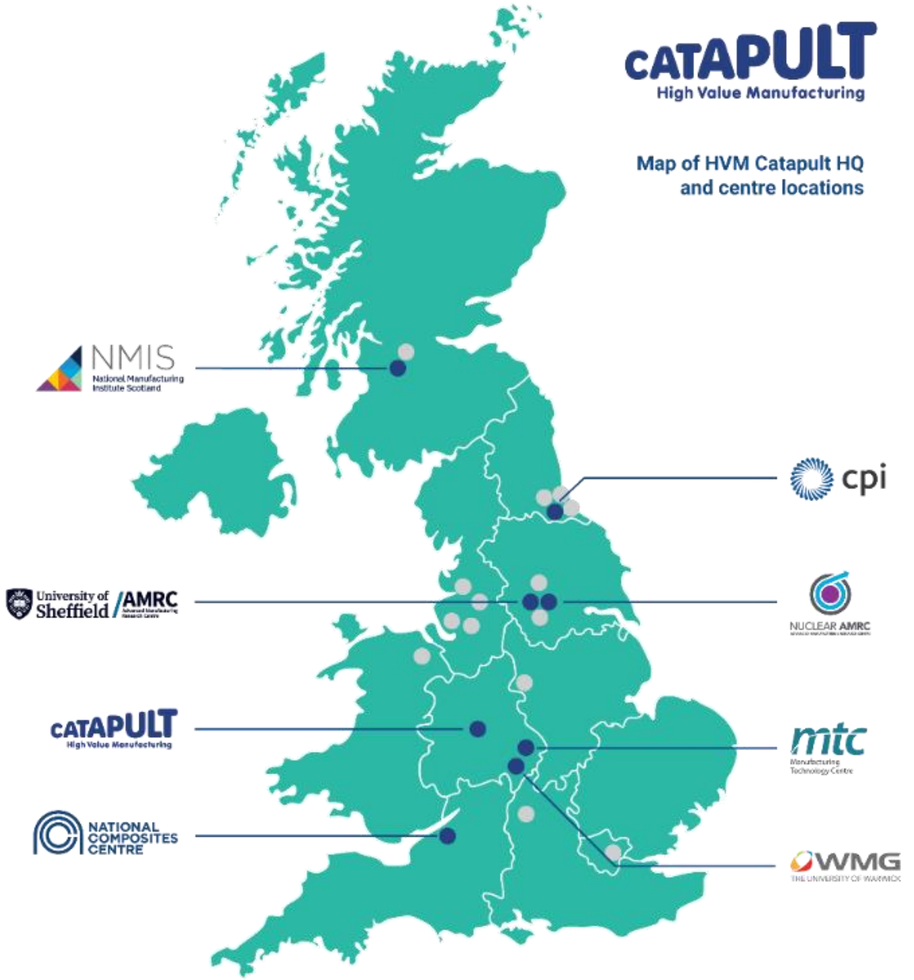
Part of the High Value Manufacturing Catapult

Established by **Innovate UK** in 2011

Seven centres of industrial innovation across 25 locations working together on the future of manufacturing in the UK.

Over 2,500 manufacturing specialists and £683m+ of state-of-the-art capability



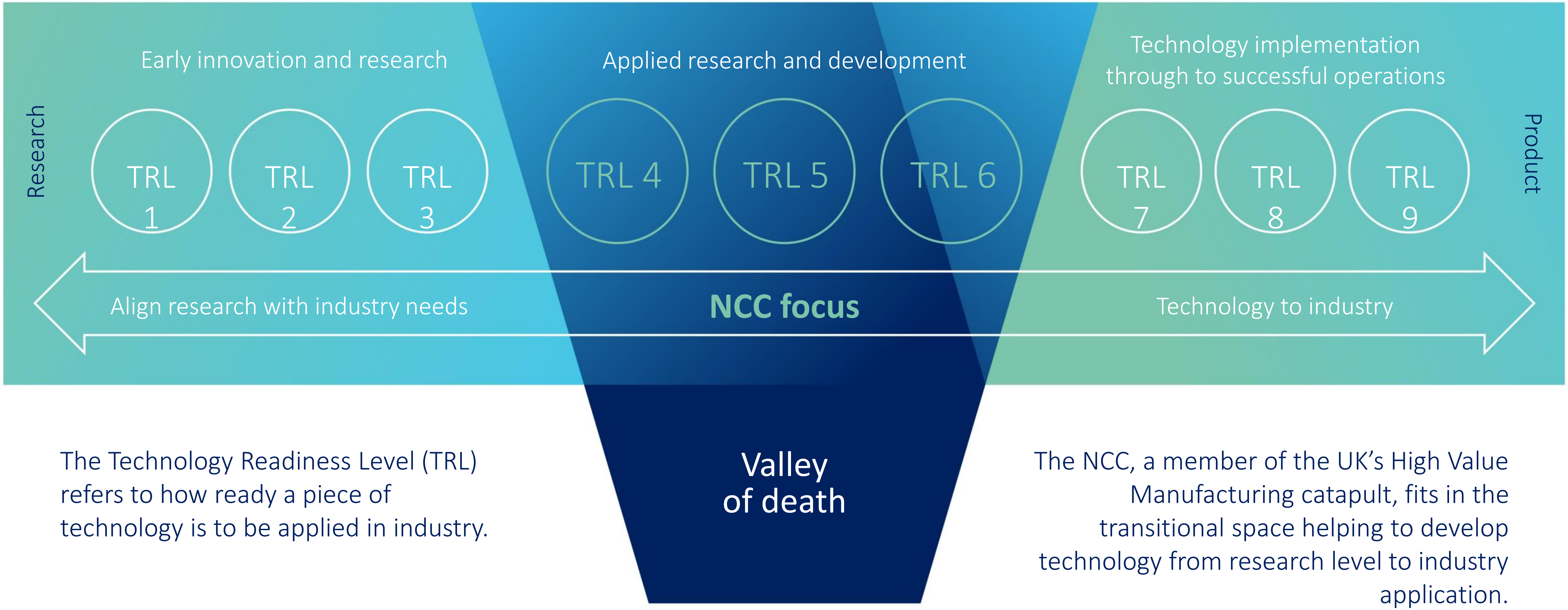
Largest advanced manufacturing capability in Europe

We've worked with more than **25,000** companies

10,000+ projects

3,500+ staff

Catapult mission: Bridging the valley of death



The Technology Readiness Level (TRL) refers to how ready a piece of technology is to be applied in industry.

The NCC, a member of the UK's High Value Manufacturing catapult, fits in the transitional space helping to develop technology from research level to industry application.

Strategic areas of focus

- National Centre of Excellence for **composite** technologies
  - Critical enabling technology for many sectors
- Using **digital engineering** to transform all aspects of product lifecycle
  - Improve time to market, productivity, in-service
- Taking the lead on **sustainability**
  - All aspects of sustainability: materials, design, reuse, recycle

Our Strategy

We are uniquely positioned to drive further industrial transformation impact

Our Vision

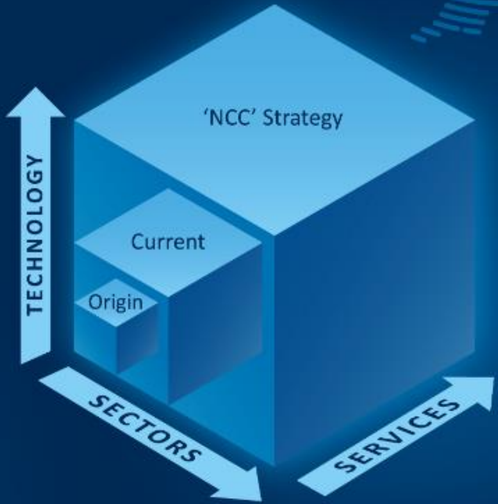
A pioneer of industrial transformation, growing our impact by delivering a range of innovations to create a sustainable, productive, and resilient future

Our Mission

Transforming Industry through collaborative innovation and accelerating its adoption

Three axes of impact and growth

Our strategy is to dramatically expand our impact along each axis



Our Strategic Priorities

Horizon 1:
Extending and expanding our established capabilities to scale up our impact in **Aerospace, Defence, Services in Sustainability, Productivity and Skills, and Systems Engineering**

Horizon 2:
Investing to build new capabilities in focus areas in **Energy Transition, Advanced Materials, Digital and AI**

Horizon 3:
Exploring our future pipeline in **other Technologies, Innovation Offerings, and our future Footprint**

Enabling:
Developing **People** and **Business Processes** to achieve our priorities in the three horizons

Impact Measures

- Gross value added per annum
- Number of products and processes improved
- Number of addressable businesses supported
- Industrial landscapes transformed
- Number of people developed
- Sustainability measurement

Early Careers Opportunities

https://www.nccuk.com/work-withus/careers/early/

NATIONAL COMPOSITES CENTRE

University of BRISTOL

CATAPULT High Value Manufacturing

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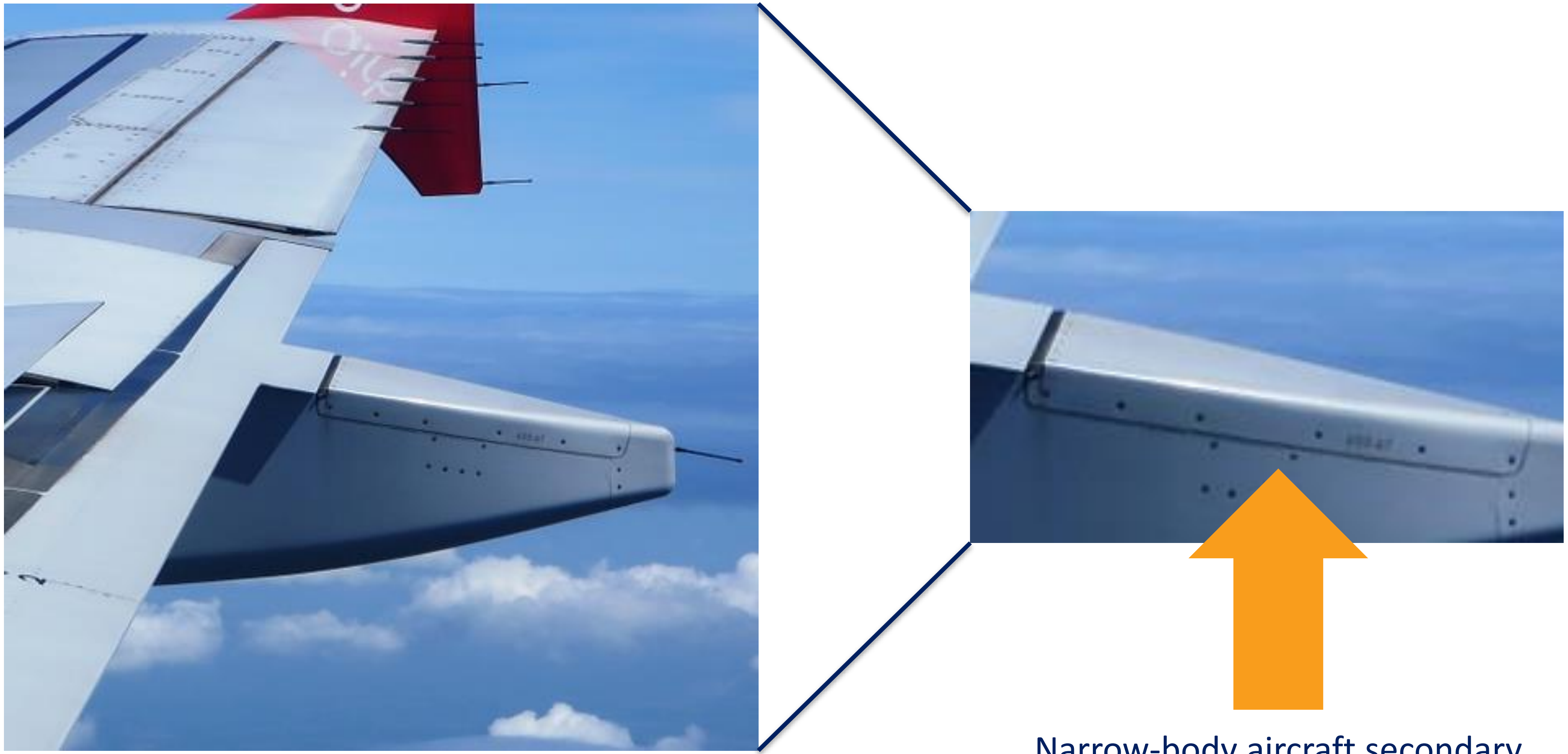
Case Study 2025 – EMS501U

Aircraft Track Fairing

The Aircraft Track Fairing

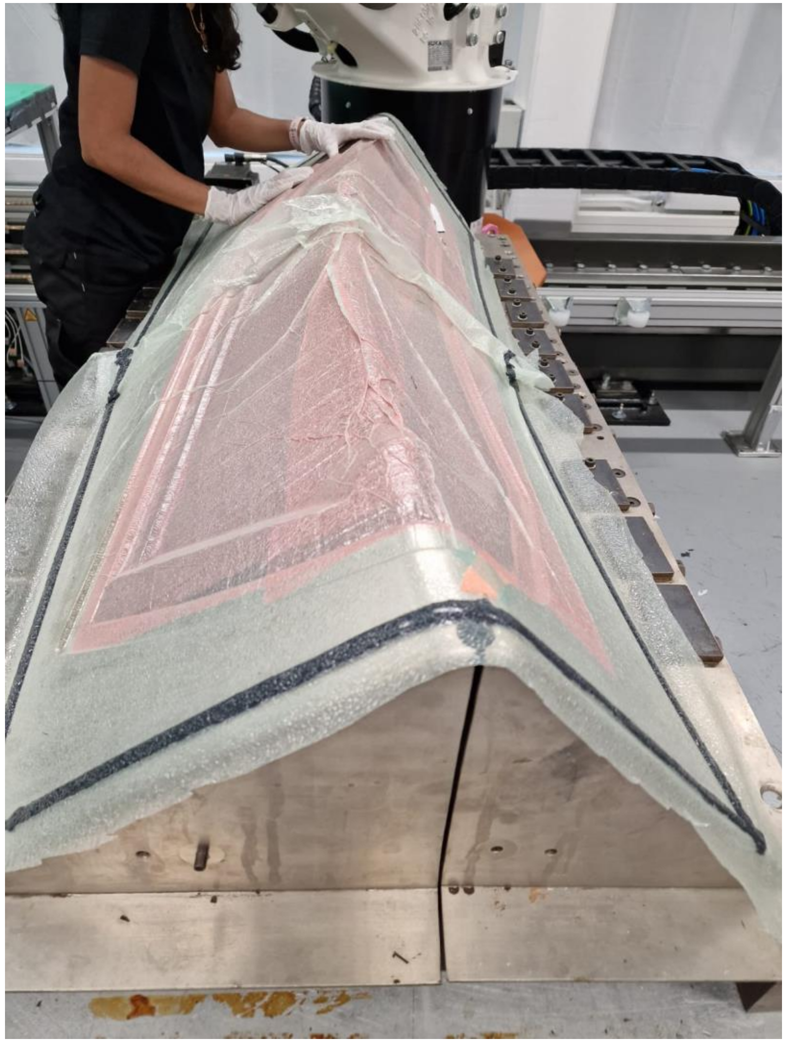


Narrow-body aircraft secondary component

The answer!

- M21E aero grade cf pre-preg specifically manufactured to meet the strict requirements for use on aircraft bodies – including, performance, safety and reg requirements (qualified materials)
- Epoxy
- Foam core: rPET
- Sandwich panel structure with ply orientated lay up
- Tool ‘on site’ at GKN (hand lay up for this secondary structure)
- Debulk with consumables
- Autoclave cure (more than one)
- Trim excess – overlay

The answer!



Why does the problem exist?

The composite manufacturing process produces un-cured offcut waste from high value material, typically 10%-35% of material purchased

Most composite manufacturing waste ends up in landfill or incineration

Value retention of waste is minimal if any

Decarbonising use-phase of products through alternative fuel sources will not hit Net Zero targets alone

Current waste recovery has low value retention

Big changes to existing systems can be complicated and costly

Many sustainable technologies are not cost effective, even at scale


CURRENT DISPOSAL ROUTE:
INCINERATION AND/OR LANDFILL

Areas for decarbonisation of aero outside new fuels

Decarbonising CF production

LIFECYCLE

Production and EOL waste

What is required to begin looking at a circular solution?

- Maximise value retention of waste fibre fabrics by retaining fibre length
- Ensure overall improvement for sustainability, considering all manufacturing steps
- Maintain or reduce any modified component’s mass
- Minimise or eliminate impact on production rates of existing components
- Be cost effective for a viable business case

How can patches help solve the problem?

Virgin material

Purchased material enters system

Primary Component

Virgin material used to manufacture primary component

Offcuts

Offcuts identified and recovered for further use

Patches

Patches nested and cut from offcut material

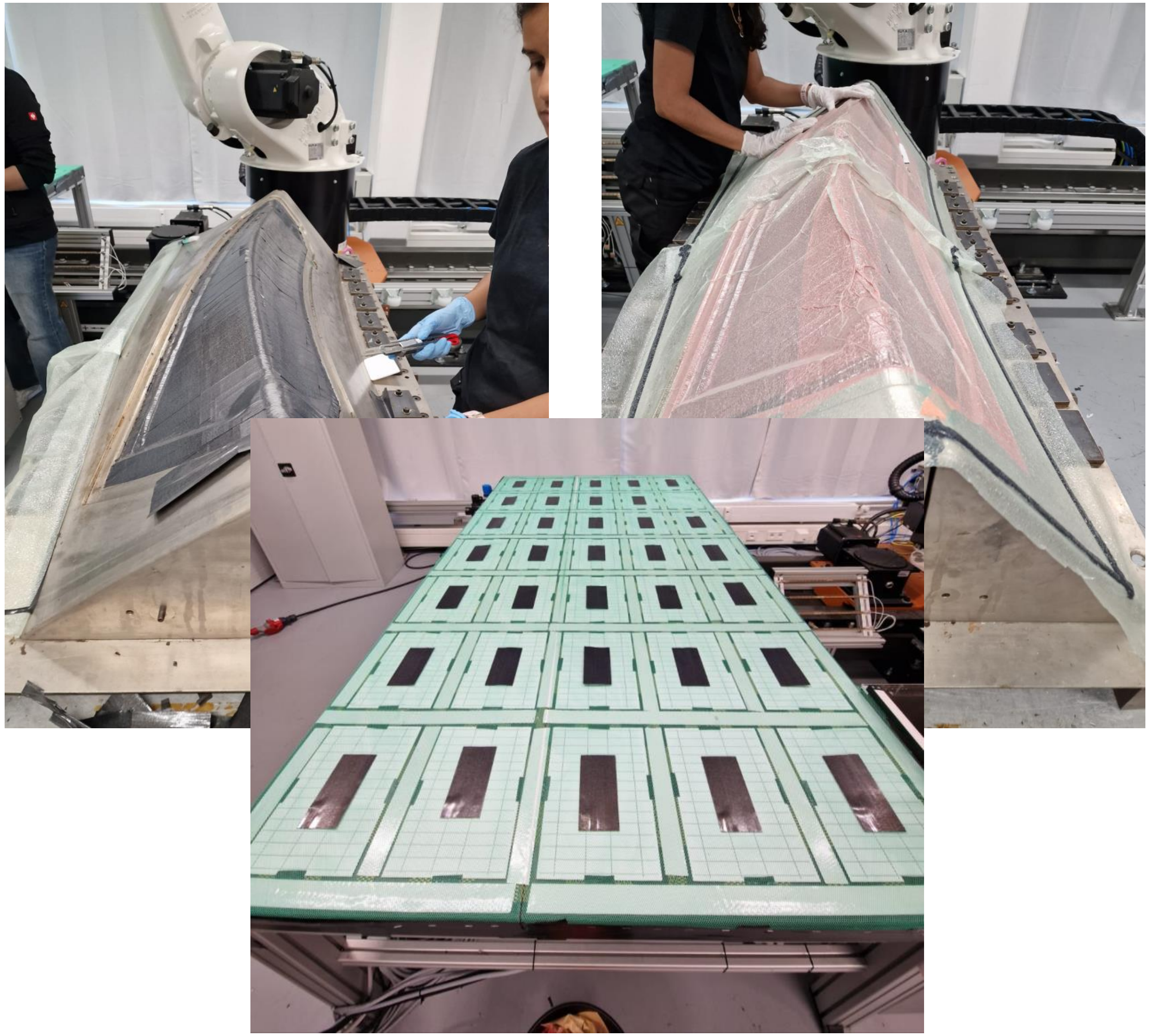
Secondary Component

Cut Patches used to manufacture secondary component

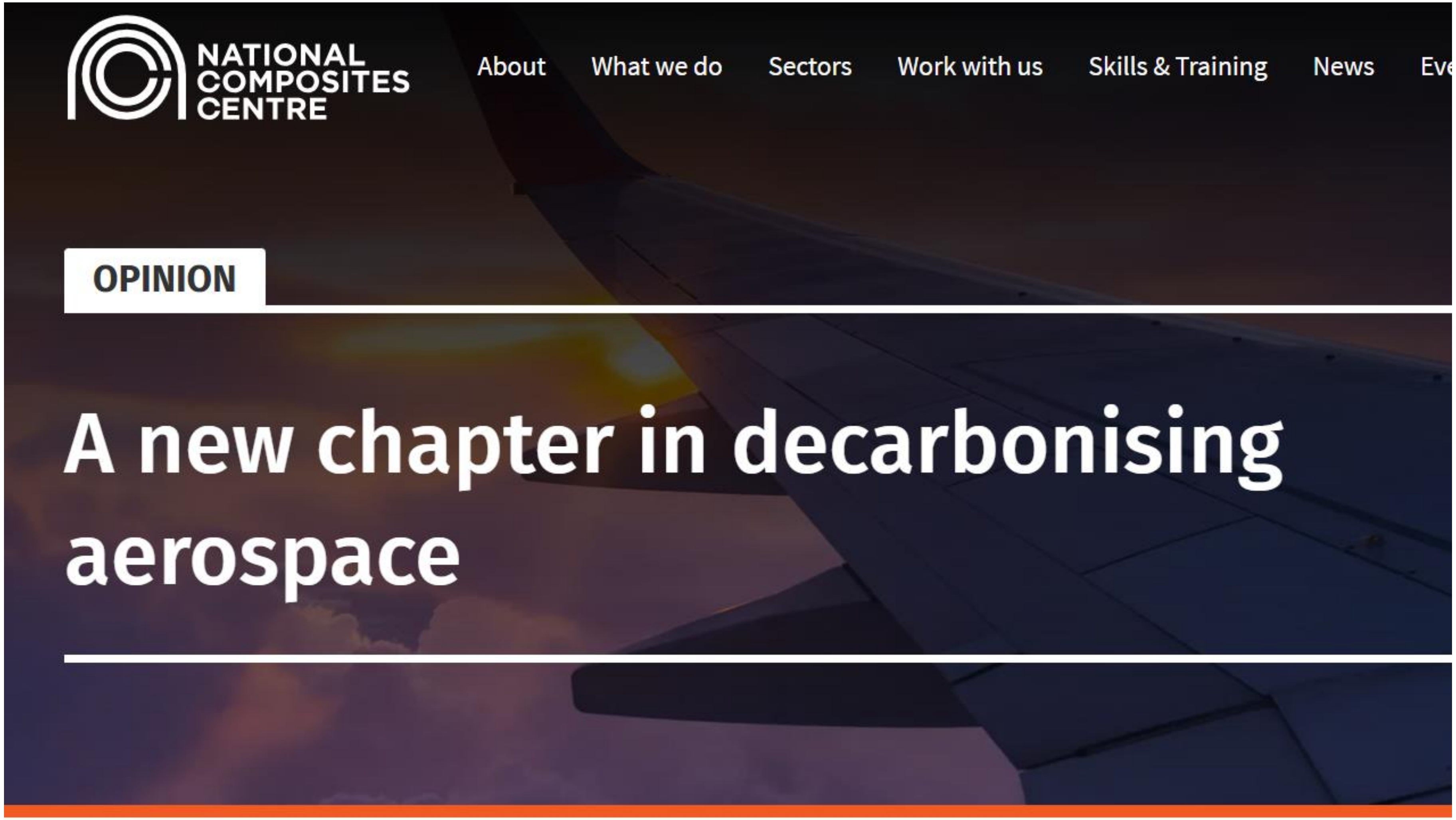
Residual Waste

Residual waste reduced, potentially used for alternative recovery

Patches to Parts



- Cevotec
- Patch size and cutting
- Logistics of prepreg storage and transportation
- Can this material be qualified as virgin rather than recycled or would it have to be redirected to replace internal glass applications – qualification difficult!



11 March 2025

Thanks!

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